

Scheme 3

Stage 3 Full-Text Deep Coding Scheme

High-resolution full-text extraction of biopsychosocial usage, factors, integration, and concepts, for the musculoskeletal and neuropathic reviews

DRAFT FOR EXPERT EVALUATION. These coding schemes are a working draft circulated for expert evaluation. They have not been applied to a final review corpus. The current manuscript is a test run that exercised an earlier, coarser generation of these schemes. The workflow itself has since been validated end to end in two cross-provider test runs, in which three large language models from three different providers applied the abstract-level and the full-text scheme independently and their agreement was quantified per coded field. The full run on the review corpus is deliberately held until this evaluation is complete.

At a Glance

Workflow position

Full-text coding after Stage 3 candidate identification and retrieval triage. It is the pass that feeds the synthesis: the domain-balance tables, the integration analysis, and the concept map.

Operational mode

One article per request, structured JSON output, deterministic vocabulary repair, item caps, and an explicit failure row when a model returns no usable coding. Nothing is fabricated to fill a gap. Human adjudication remains the final authority on eligibility, as the registration specifies.

Unit of analysis

One retrieved full-text review, coded against the complete text. Every extracted item carries a verbatim quote and the section it came from.

Evidence rule

Quotes are checked against the source text after the run. A quote that cannot be found in the article is reported as unverified rather than accepted.

Resolution

Thirteen structured extraction lists, seven open free-text lists, 82 fields inside the list items, and a ceiling of 116 extracted items per coding. Presence is never the answer; the answer is which ones.

Provenance basis

The implementation in bps_review/fulltext (schema, vocabulary, prompt, derivations, runner) and the human coding template generated by stage3_prep.py.

Research questions

RQ1 (what the BPS label actually does, passage by passage); RQ2 (scope, balance, integration, with named factors on both ends of every link); RQ3 (concepts, definitions, hierarchical and semantic relations, frameworks, measures); SQ1 (conceptual problems, with the constructs they concern)

Tagline

Graded ladders with a quoted passage behind every rung, plus a named-item extraction layer that supplies the nodes and edges of the biopsychosocial ontology

Canonical Source Paths

- `src/03_pipeline/bps_review/fulltext/coding/schema.py`
- `src/03_pipeline/bps_review/fulltext/coding/vocabulary.py`
- `src/03_pipeline/bps_review/fulltext/coding/prompt.py`
- `src/03_pipeline/bps_review/fulltext/coding/derive.py`
- `src/03_pipeline/bps_review/fulltext/coding/runner.py`
- `src/03_pipeline/bps_review/fulltext/analysis/integrity.py`
- `src/01_protocol/codebooks/stage3_codebook.md`
- `src/09_review_stages/04_extraction/codebooks/stage3_codebook.csv`
- `src/09_review_stages/04_extraction/forms/stage3_fulltext_coding_template.csv`

Purpose

This scheme is the full-text deep coding framework for Stage 3 candidate reviews. It is applied as one uniform instrument to both planned reviews: the musculoskeletal chronic pain review and the neuropathic chronic pain review. The pain-condition family is the varying input that decides which records each review reads; the coding fields, value vocabularies, and anchors are identical across both tracks so the two reviews stay directly comparable.

It does two things at once. It grades: how deeply each domain is treated, and how each pair of domains and the triad are integrated, on explicit ladders with a quoted passage behind every rung. And it extracts: the specific things a review names. The extraction half is the larger one. Recording that a domain is present is not a coding; recording which biological, psychological, social, lifestyle, and existential factors carry it, what role each plays, and on the strength of which sentence, is.

Resolution is the point. A single article routinely yields between thirty and seventy extracted items: passages where the biopsychosocial label does work, factors per domain, psychological constructs with their definitional status, relations drawn between constructs, frameworks, instruments, conceptual problems, and quotable claims. Those items are the nodes and edges of the biopsychosocial ontology the synthesis builds, so they have to be nameable, countable, and traceable to a passage.

Every extraction list carries a maximum number of items. The cap is a ceiling and never a target: a list is left empty when the article offers nothing of that kind, and an empty list is itself a coding, not a gap.

Stage 3 is where the review's central claim is tested: does a BPS-labelled review actually integrate the three domains, and if so, how, between which factors, and by what mechanism.

Carried-Through Metadata Fields

Descriptive fields retained from earlier stages and from retrieval, not coded by the model.

- `record_id`, `abstract_record_id`, `pmid`, `pmcid`, `doi`
- `title`, `journal`, `year`, `authors`
- `n_sections`, `section_titles`, `body_chars` (what was available to read)
- `coder_id`, `coding_method`, `llm_model`, `adjudication_status` (provenance of the row)

A. Source, Context, and Routing

What kind of source this is, which review it belongs to, and which population, condition, and setting it concerns. These are the registration's contextual variables, now read from the full text rather than inferred from an abstract.

review_track

Which review this coded record belongs to. The coding fields are uniform across both; the track only tunes which biological ontology extension the coder reads the biological domain against.

- **musculoskeletal** Low back, neck, osteoarthritis, fibromyalgia, shoulder, and similar.
- **neuropathic** Painful neuropathy, radicular pain, post-herpetic neuralgia, and similar.
- **mixed_or_other** Several pain families are genuinely covered, or the condition belongs to neither.
- **unclear** The paper does not say.

source_type

The evidence-synthesis design, read from how the paper describes itself in its abstract and methods.

- **systematic review** Explicit systematic methods.
- **meta-analysis** Quantitative pooling of effect sizes. *Boundary:* Outranks systematic review when pooling is explicit.
- **network meta-analysis** Multiple-treatment comparison with indirect evidence.
- **umbrella review** Review of reviews.
- **scoping or mapping review** Breadth-oriented mapping.
- **rapid review** Streamlined systematic methods.
- **realist review** Theory-driven mechanism review.
- **integrative review** Mixed evidence integration.
- **narrative or expert review** Non-systematic expert synthesis.
- **clinical guideline or consensus statement** A guideline or a formal consensus.
- **other evidence synthesis** A synthesis type none of the values above describes.
- **primary study** Reads as a primary study rather than a synthesis. *Boundary:* An eligibility signal: the registration excludes primary studies, so this routes the record to human adjudication.
- **unclear** The design cannot be determined.

icd11_pain_category

The ICD-11 aligned pain category, now read from the full text. Recording it again at Stage 3 makes the abstract-level classification checkable rather than final.

- **chronic secondary musculoskeletal pain** Low back, neck, osteoarthritis, whiplash, and similar.
- **chronic neuropathic pain** Neuropathic, CRPS, radiculopathy.
- **chronic cancer-related pain** Cancer or oncology pain.
- **chronic postsurgical or posttraumatic pain** Persistent pain after surgery or trauma.
- **chronic secondary headache or orofacial pain** Headache, migraine, orofacial, TMD.
- **chronic secondary visceral pain** Visceral, abdominal, pelvic pain.

- **chronic primary pain** Fibromyalgia and other primary pain syndromes.
- **mixed or unspecified chronic pain** Several categories genuinely covered.
- **unclear** The category cannot be inferred.

population

The population the reviewed evidence concerns.

- **adult** Adults, the eligible population.
- **older adult** Explicitly an older population, foregrounded by the paper.
- **mixed ages** Adults and younger participants both included.
- **pediatric** Children or adolescents only. *Boundary:* An exclusion signal under the registration.
- **unclear** The paper does not report it.
- **not applicable** A purely theoretical paper with no population.

care_setting

The care setting the paper concerns, when it reports one. The registration lists setting among the contextual variables to extract.

- **primary care** General practice and first-line care.
- **secondary or tertiary specialist care** Specialist clinics, pain centres, hospital care.
- **rehabilitation or multidisciplinary programme** Rehabilitation and multimodal programmes.
- **occupational or workplace** Workplace and occupational health settings.
- **community or population** Community, population, or public-health settings.
- **mixed** Several settings genuinely covered.
- **not reported** The paper does not say. *Boundary:* The honest answer for most reviews, and preferred over a guess.

primary_discipline

The disciplinary home of the paper, read from the journal, the framing, and the vocabulary rather than from author affiliations alone.

- **physiotherapy or rehabilitation**
- **clinical or health psychology**
- **rheumatology or orthopaedics**
- **pain medicine or anaesthesiology**
- **neurology or neuroscience**
- **nursing**
- **general or family medicine**
- **public health or epidemiology**
- **multidisciplinary** The paper is genuinely written across disciplines. *Boundary:* Multidisciplinary describes the writing, not the author list.
- **other** A discipline none of the values above describes.
- **unclear** It cannot be read off the paper.

Discipline is a covariate for the synthesis: whether biopsychosocial usage differs by field is one of the descriptive questions the corpus can answer.

pain_condition_detail

The exact pain condition or conditions studied, in the paper's own words. At most 40 words. (*free text*)

pain_conditions

The specific pain conditions named, as a list. Preferred labels exist (chronic low back pain, knee osteoarthritis, painful diabetic neuropathy, and others) and the paper's own wording is used whenever it is more precise. (*open list; max 6 items; free text*)

context_note

The cultural, geographic, or healthcare-system context, when the paper states one. At most 40 words, empty when it does not. (*free text*)

quality_assessment_reported

Whether the paper reports a formal quality or risk-of-bias assessment of the evidence it reviews.

- **yes** An appraisal is reported.
- **no** No appraisal is reported.
- **unclear** It cannot be determined.

Descriptive only. The registration is explicit that this review does not appraise the methodological quality of the reviews it studies.

quality_assessment_tools

The appraisal tools named (AMSTAR, AMSTAR-2, ROBIS, GRADE, Cochrane risk of bias, and others). Empty when none is named. (*open list; max 4 items; free text*)

B. What the Biopsychosocial Label Does (RQ1)

The primary research question of the review, coded at passage level. The registration asks for the location and the function of the biopsychosocial mention; at full text that becomes an inventory, because one paper routinely uses the label for two or three different purposes in different sections, and which purposes it combines is the finding.

bps_label_used

Which biopsychosocial vocabulary the paper actually uses.

- **explicit_bps_term** The words biopsychosocial or bio-psycho-social appear somewhere in the text.
- **variant_term_only** Only a neighbouring term appears (psychosocial, multidimensional, multifactorial, holistic). *Boundary:* Prefer `explicit_bps_term` whenever the full term appears at all, even once.
- **domain_language_only** The domains are discussed with no model label at all.
- **absent** Neither the label nor domain language appears.

bps_primary_function

The single dominant work the label does, judged over the paper as a whole. Shares its vocabulary with the abstract-level scheme so the two readings of a record are directly comparable.

- **explanatory framework** BPS is used as a model that explains pain or pain-related disability. *Choose when:* explains, accounts for, mechanism of. *Not this if:* Only justifies multimodal treatment, which is intervention rationale. *Boundary:* Requires explanatory intent, not only endorsement of the model's existence.
- **intervention rationale** BPS mainly justifies multimodal or interdisciplinary treatment.
- **organizing principle** BPS structures the scope or the categories of the review without specifying integration mechanisms. *Choose when:* We organize factors into bio, psycho, social. *Not this if:* A stated cross-domain mechanism.
- **justification** BPS justifies the relevance or the importance of the topic.
- **background framing** BPS sets context in the background without analytic use.
- **conclusion** BPS appears mainly in the concluding claims.
- **policy or practice implication** BPS frames a policy or practice recommendation.
- **rhetorical label** BPS is invoked ceremonially, aspirationally, or symbolically with no substantive analytic work. *Choose when:* a biopsychosocial approach is needed, with no follow-through.
- **critique or problematization** The paper argues about the model itself: what it leaves out, what it cannot explain, how it is misused. *Boundary:* New at full-text level. These papers are the most informative for the review and are almost invisible in an abstract.
- **operational definition** The paper turns the model into the variables it actually codes or measures. *Boundary:* New at full-text level. Reserve it for a paper that operationalizes the model, not one that merely organizes prose by domain.
- **unclear** The function cannot be determined.

bps_functions_present

Every function the label performs anywhere in the paper, as a multi-label list from the same vocabulary. A paper routinely does two or three of these at once. (*multi-label list; max 6 items*)

- **explanatory framework**
- **intervention rationale**
- **organizing principle**
- **justification**
- **background framing**
- **conclusion**
- **policy or practice implication**
- **rhetorical label**
- **critique or problematization**
- **operational definition**
- **unclear**

This is the field that answers RQ1 as a distribution rather than as a single label. A paper's usage collapsed into one function loses the pattern the review is looking for.

bps_definition_status

How the paper handles the meaning of the model itself.

- **formally_defined** The paper states what the model means.

- **described_informally** The meaning is carried by description rather than by a definition.
- **cited_only** A citation stands in for a definition.
- **undefined** The label is used with no meaning given anywhere. *Boundary:* A finding, not a coding failure. Papers that use the model without ever saying what it is are exactly what this review is about.

bps_model_variants

The model labels the paper actually uses, verbatim and de-duplicated (biopsychosocial model, biopsychosocial framework, sociopsychobiological model, extended biopsychosocial model, and others). This is what makes terminological drift visible. (*open list; max 5 items; free text*)

bps_usage_instances

One item for every distinct passage where the label does work. A paper that invokes the model in the introduction to justify the topic and again in the discussion to recommend multidisciplinary care yields two items, not one. (*extraction list; max 8 items*) *Fields inside each item:*

- **usage_verbatim** The exact passage, at most 60 words.
- **bps_function** The function the label serves in this passage. *Values:* explanatory framework, intervention rationale, organizing principle, justification, background framing, conclusion, policy or practice implication, rhetorical label, critique or problematization, operational definition, unclear
- **is_definitional** Whether this passage also says what the model is. *Values:* yes, no
- **attributed_source** Who the model is credited to here (Engel, Gatchel, Waddell, IASP, a guideline, or nobody).
- **section_located** Where the passage appears. *Values:* abstract, introduction, methods, results, discussion, conclusion, table or figure, other, unclear
- **note** Anything the fields above cannot hold.

bps_definitions

One item for every place where the paper says what the biopsychosocial model is. Empty when it never does. (*extraction list; max 3 items*) *Fields inside each item:*

- **definition_verbatim** The exact passage, at most 60 words.
- **definition_type** What kind of definitional act it is. *Values:* explicit_formal, operational, implicit_description, borrowed, critique_of_definition, other
- **attributed_source** The citation the paper credits.
- **elements_named** The components the definition lists, as short labels.
- **section_located** Where the passage appears. *Values:* abstract, introduction, methods, results, discussion, conclusion, table or figure, other, unclear

bps_operationalization_summary

At most 90 words, in the coder's words: what this paper actually does with the model, as opposed to what it says about it. Written to name the mechanism of use, for example organizes the results section into three domain headings and never relates them. (*free text*)

C. Domain Coverage

Coverage is coded at the level of substantive treatment, not keyword presence. The same four-level ladder applies to each domain and is identical across both reviews. Only the biological reading is track-aware: for a musculoskeletal record the biological content is judged against musculoskeletal mechanisms, and for a neuropathic record against neuropathic mechanisms (see Scheme 6).

domain_coverage_bio

Depth of biological content, read against the track-appropriate biological mechanisms.

- **elaborated** The domain is developed as a substantive analytic thread. Several distinct constructs or mechanisms are discussed, weighed, or synthesized rather than merely named. *Choose when:* A dedicated section or sustained multi-paragraph treatment; Two or more distinct constructs named and related to one another; Evidence is appraised, not only cited in passing. *Not this if:* A single passing sentence; Only the domain umbrella word. *Boundary:* Prefer elaborated over mentioned when the domain carries its own argument rather than appearing as one item in a list.
- **mentioned** The domain is explicitly present with at least one named construct, but it is not developed into a sustained analytic thread. *Choose when:* One or two named constructs; A short paragraph or scattered sentences. *Not this if:* Only the umbrella word such as the bare token psychosocial. *Boundary:* Prefer mentioned over minimal when a concrete construct is named rather than only the domain umbrella term.
- **minimal** The domain appears only as an umbrella label or a single incidental reference, with no concrete construct attached. *Choose when:* Bare token such as social factors with no elaboration; One incidental clause. *Not this if:* Any named construct, which would qualify as mentioned. *Boundary:* Prefer minimal over absent when the domain word appears at all.
- **absent** The domain is not represented in the substantive content of the review, even if a global BPS label is used elsewhere. *Choose when:* No domain-specific language beyond a global BPS label. *Not this if:* Any named construct or umbrella mention. *Boundary:* A review can carry a BPS label and still score a domain as absent. The label alone is never counted as coverage.

domain_coverage_psych

Depth of psychological content.

- **elaborated** The domain is developed as a substantive analytic thread. Several distinct constructs or mechanisms are discussed, weighed, or synthesized rather than merely named. *Choose when:* A dedicated section or sustained multi-paragraph treatment; Two or more distinct constructs named and related to one another; Evidence is appraised, not only cited in passing. *Not this if:* A single passing sentence; Only the domain umbrella word. *Boundary:* Prefer elaborated over mentioned when the domain carries its own argument rather than appearing as one item in a list.
- **mentioned** The domain is explicitly present with at least one named construct, but it is not developed into a sustained analytic thread. *Choose when:* One or two named constructs; A short paragraph or scattered sentences. *Not this if:* Only the umbrella word such as the bare token psychosocial. *Boundary:* Prefer mentioned over minimal when a concrete construct is named rather than only the domain umbrella term.
- **minimal** The domain appears only as an umbrella label or a single incidental reference, with no concrete construct attached. *Choose when:* Bare token such as social factors with no elaboration; One incidental clause. *Not this if:* Any named construct, which would qualify as mentioned. *Boundary:* Prefer minimal over absent when the domain word appears at all.
- **absent** The domain is not represented in the substantive content of the review, even if a global BPS label is used elsewhere. *Choose when:* No domain-specific language beyond a global BPS label. *Not this if:* Any named construct or umbrella mention. *Boundary:* A review can carry a BPS label and still score a domain as absent. The label alone is never counted as coverage.

domain_coverage_social

Depth of social content.

- **elaborated** The domain is developed as a substantive analytic thread. Several distinct constructs or mechanisms are discussed, weighed, or synthesized rather than merely named. *Choose when:* A dedicated section or sustained multi-paragraph treatment; Two or more distinct constructs named and related to one another; Evidence is appraised, not only cited in passing. *Not this if:* A single passing sentence; Only the domain umbrella word. *Boundary:* Prefer elaborated over mentioned when the domain carries its own argument rather than appearing as one item in a list.
- **mentioned** The domain is explicitly present with at least one named construct, but it is not developed into a sustained analytic thread. *Choose when:* One or two named constructs; A short paragraph or scattered sentences. *Not this if:* Only the umbrella word such as the bare token psychosocial. *Boundary:* Prefer mentioned over minimal when a concrete construct is named rather than only the domain umbrella term.
- **minimal** The domain appears only as an umbrella label or a single incidental reference, with no concrete construct attached. *Choose when:* Bare token such as social factors with no elaboration; One incidental clause. *Not this if:* Any named construct, which would qualify as mentioned. *Boundary:* Prefer minimal over absent when the domain word appears at all.
- **absent** The domain is not represented in the substantive content of the review, even if a global BPS label is used elsewhere. *Choose when:* No domain-specific language beyond a global BPS label. *Not this if:* Any named construct or umbrella mention. *Boundary:* A review can carry a BPS label and still score a domain as absent. The label alone is never counted as coverage.

coverage_lifestyle

Depth of lifestyle content on the same ladder: physical activity and exercise behaviour, sleep hygiene, diet and weight, smoking, alcohol.

- **elaborated** The domain is developed as a substantive analytic thread. Several distinct constructs or mechanisms are discussed, weighed, or synthesized rather than merely named. *Choose when:* A dedicated section or sustained multi-paragraph treatment; Two or more distinct constructs named and related to one another; Evidence is appraised, not only cited in passing. *Not this if:* A single passing sentence; Only the domain umbrella word. *Boundary:* Prefer elaborated over mentioned when the domain carries its own argument rather than appearing as one item in a list.
- **mentioned** The domain is explicitly present with at least one named construct, but it is not developed into a sustained analytic thread. *Choose when:* One or two named constructs; A short paragraph or scattered sentences. *Not this if:* Only the umbrella word such as the bare token psychosocial. *Boundary:* Prefer mentioned over minimal when a concrete construct is named rather than only the domain umbrella term.
- **minimal** The domain appears only as an umbrella label or a single incidental reference, with no concrete construct attached. *Choose when:* Bare token such as social factors with no elaboration; One incidental clause. *Not this if:* Any named construct, which would qualify as mentioned. *Boundary:* Prefer minimal over absent when the domain word appears at all.
- **absent** The domain is not represented in the substantive content of the review, even if a global BPS label is used elsewhere. *Choose when:* No domain-specific language beyond a global BPS label. *Not this if:* Any named construct or umbrella mention. *Boundary:* A review can carry a BPS label and still score a domain as absent. The label alone is never counted as coverage.

The registration lists lifestyle factors alongside the three core domains. Coding it separately keeps it out of the triad, where it would inflate biological or social coverage.

coverage_spiritual_existential

Depth of spiritual or existential content on the same ladder: meaning, faith or religion, hope, existential suffering.

- **elaborated** The domain is developed as a substantive analytic thread. Several distinct constructs or mechanisms are discussed, weighed, or synthesized rather than merely named. *Choose when:* A dedicated section or sustained multi-paragraph treatment; Two or more distinct constructs named and related to one another; Evidence is appraised, not only cited in passing. *Not this if:* A single passing sentence; Only the domain umbrella word. *Boundary:* Prefer elaborated over mentioned when the domain carries its own argument rather than appearing as one item in a list.
- **mentioned** The domain is explicitly present with at least one named construct, but it is not developed into a sustained analytic thread. *Choose when:* One or two named constructs; A short paragraph or scattered sentences. *Not this if:* Only the umbrella word such as the bare token psychosocial. *Boundary:* Prefer mentioned over minimal when a concrete construct is named rather than only the domain umbrella term.
- **minimal** The domain appears only as an umbrella label or a single incidental reference, with no concrete construct attached. *Choose when:* Bare token such as social factors with no elaboration; One incidental clause. *Not this if:* Any named construct, which would qualify as mentioned. *Boundary:* Prefer minimal over absent when the domain word appears at all.
- **absent** The domain is not represented in the substantive content of the review, even if a global BPS label is used elsewhere. *Choose when:* No domain-specific language beyond a global BPS label. *Not this if:* Any named construct or umbrella mention. *Boundary:* A review can carry a BPS label and still score a domain as absent. The label alone is never counted as coverage.

Also registered as a domain of its own. Absent is the expected value for most papers, and that distribution is itself a result.

domain_evidence

One item per core domain not scored as absent, carrying the passage that justifies the coverage level. (*extraction list; max 5 items*) *Fields inside each item:*

- **domain** Which domain this passage carries. *Values:* biological, psychological, social
- **coverage_level** The level given above, repeated here so the judgement and its evidence travel together. *Values:* elaborated, mentioned, minimal, absent
- **constructs_named** The domain-specific constructs the paper actually names.
- **subdomains_named** The Scheme 6 ontology subdomains the content belongs to, mapped when they fit.
- **evidence_verbatim** The passage, at most 60 words.
- **section_located** Where the passage appears. *Values:* abstract, introduction, methods, results, discussion, conclusion, table or figure, other, unclear

D. Which Factors Carry Each Domain (the Ontology Nodes)

The core of the extraction layer. A coverage grade says how much of a domain a review carries; these lists say what it carries. Each item holds the review's own label, an anchor onto the project ontology where one fits, the role the factor plays in the paper's account, and the passage behind it. Psychological constructs are recorded in section G instead, which carries the extra definitional fields RQ3 needs, so no factor is ever written twice.

biological_factors

Every biological factor the paper names. (*extraction list; max 12 items*) *Fields inside each item:*

- **factor_label** The paper's own term, as specifically as the paper puts it.
- **subdomain_label** The Scheme 6 biological subdomain it belongs to, when one fits. Empty when none does, and the label then also goes into emergent_labels.
- **mechanism_level** Where the factor sits. *Values:* peripheral or tissue, spinal or central nervous system, systemic or whole body, genetic or molecular, structural or anatomical, treatment related, other, unclear
- **factor_role** What the factor does in this paper. *Values:* determinant or risk factor, protective factor, mediator, moderator, outcome, correlate, treatment target, intervention component, contextual condition, descriptive theme, other, unclear
- **factor_verbatim** The passage, at most 60 words.
- **section_located** Where the passage appears. *Values:* abstract, introduction, methods, results, discussion, conclusion, table or figure, other, unclear
- **evidence_basis** What the claim rests on. *Values:* asserted, theorized, empirically_supported, empirically_contested, cited_from_other_work, clinical_observation, other, unclear

The role field is what separates a review that lists biology from one that gives biology a job. A factor coded as mediator or moderator is a different contribution from the same factor coded as a descriptive theme.

social_factors

Every social factor the paper names, with the level of social organization it sits at. (*extraction list; max 12 items*) *Fields inside each item:*

- **factor_label** The paper's own term.
- **subdomain_label** The Scheme 6 social subdomain, when one fits.
- **social_level** The level of social organization. *Values:* interpersonal, family or household, workplace, community, healthcare system, societal or policy, cultural, economic, other, unclear
- **factor_role** What the factor does in this paper. *Values:* determinant or risk factor, protective factor, mediator, moderator, outcome, correlate, treatment target, intervention component, contextual condition, descriptive theme, other, unclear
- **factor_verbatim** The passage, at most 60 words.
- **section_located** Where the passage appears. *Values:* abstract, introduction, methods, results, discussion, conclusion, table or figure, other, unclear
- **evidence_basis** What the claim rests on. *Values:* asserted, theorized, empirically_supported, empirically_contested, cited_from_other_work, clinical_observation, other, unclear

The social domain is the one this literature is thinnest on, and the level field is what shows whether social means an interpersonal relationship or a structural condition. Both are coded as social by the coverage ladder; they are not the same claim.

other_domain_factors

Factors outside the triad, kept visible rather than folded into it. (*extraction list; max 6 items*) *Fields inside each item:*

- **factor_label** The paper's own term.
- **domain** Which non-triad domain it belongs to. *Values:* lifestyle, spiritual or existential, environmental, other

- **factor_role** What the factor does in this paper. *Values:* determinant or risk factor, protective factor, mediator, moderator, outcome, correlate, treatment target, intervention component, contextual condition, descriptive theme, other, unclear
- **factor_verbatim** The passage, at most 60 words.
- **section_located** Where the passage appears. *Values:* abstract, introduction, methods, results, discussion, conclusion, table or figure, other, unclear

E. Integration (the Core RQ2 Contribution, the Ontology Edges)

Integration is the scheme's highest-resolution construct. The pairwise ladder distinguishes a stated mechanism from a mere direction, an association, or a bare co-mention, and every graded link has to point at the sentence that carries it. The claim items now name the two factors on either end of the link, which is what turns a count of integration statements into a map of what this literature says connects to what.

integration_bio_psych

Biological to psychological integration.

- **mechanistic** The review specifies a pathway or mechanism by which one domain acts on the other (a how, not only a that). *Choose when:* Named mediator, moderator, or physiological or behavioural pathway; Language such as via, through, drives, sensitizes, amplifies. *Not this if:* Two domains listed as co-occurring with no linking process. *Boundary:* Mechanistic requires a stated process; if only a direction of effect is claimed without a pathway, use directional.
- **directional** A directional or causal influence between the two domains is asserted, but no mechanism is given. *Choose when:* X predicts, increases, or worsens Y; A stated arrow of influence. *Not this if:* A pathway is described, which would be mechanistic; Mere co-mention with no direction. *Boundary:* Directional requires an arrow of effect; if the link is only an association or correlation, use descriptive.
- **descriptive** The two domains are linked as associated or correlated, without direction or mechanism. *Choose when:* Associated with, related to, correlated with, linked to. *Not this if:* A stated direction of effect, which would be directional. *Boundary:* Descriptive requires an explicit relational claim; a bare co-occurrence in the same sentence is only mentioned.
- **mentioned** Both domains appear near one another but the relationship is not characterized at all. *Choose when:* Two domains named in the same passage with no relational verb. *Not this if:* Any associational, directional, or mechanistic claim. *Boundary:* Prefer mentioned over none only when both domains are present in a shared context rather than in separate isolated sections.
- **none** No relationship between the two domains is articulated anywhere in the review. *Choose when:* Domains handled in separate silos, or one domain absent. *Not this if:* Any co-located relational statement. *Boundary:* If either domain is absent, the pairwise code defaults to none.

integration_psych_social

Psychological to social integration.

- **mechanistic** The review specifies a pathway or mechanism by which one domain acts on the other (a how, not only a that). *Choose when:* Named mediator, moderator, or physiological or behavioural pathway; Language such as via, through, drives, sensitizes, amplifies. *Not this if:* Two domains listed as co-occurring with no linking process. *Boundary:* Mechanistic requires a stated process; if only a direction of effect is claimed without a pathway, use directional.
- **directional** A directional or causal influence between the two domains is asserted, but no mechanism is given. *Choose when:* X predicts, increases, or worsens Y; A stated arrow of influence.

Not this if: A pathway is described, which would be mechanistic; Mere co-mention with no direction. *Boundary:* Directional requires an arrow of effect; if the link is only an association or correlation, use descriptive.

- **descriptive** The two domains are linked as associated or correlated, without direction or mechanism. *Choose when:* Associated with, related to, correlated with, linked to. *Not this if:* A stated direction of effect, which would be directional. *Boundary:* Descriptive requires an explicit relational claim; a bare co-occurrence in the same sentence is only mentioned.
- **mentioned** Both domains appear near one another but the relationship is not characterized at all. *Choose when:* Two domains named in the same passage with no relational verb. *Not this if:* Any associational, directional, or mechanistic claim. *Boundary:* Prefer mentioned over none only when both domains are present in a shared context rather than in separate isolated sections.
- **none** No relationship between the two domains is articulated anywhere in the review. *Choose when:* Domains handled in separate silos, or one domain absent. *Not this if:* Any co-located relational statement. *Boundary:* If either domain is absent, the pairwise code defaults to none.

integration_bio_social

Biological to social integration.

- **mechanistic** The review specifies a pathway or mechanism by which one domain acts on the other (a how, not only a that). *Choose when:* Named mediator, moderator, or physiological or behavioural pathway; Language such as via, through, drives, sensitizes, amplifies. *Not this if:* Two domains listed as co-occurring with no linking process. *Boundary:* Mechanistic requires a stated process; if only a direction of effect is claimed without a pathway, use directional.
- **directional** A directional or causal influence between the two domains is asserted, but no mechanism is given. *Choose when:* X predicts, increases, or worsens Y; A stated arrow of influence. *Not this if:* A pathway is described, which would be mechanistic; Mere co-mention with no direction. *Boundary:* Directional requires an arrow of effect; if the link is only an association or correlation, use descriptive.
- **descriptive** The two domains are linked as associated or correlated, without direction or mechanism. *Choose when:* Associated with, related to, correlated with, linked to. *Not this if:* A stated direction of effect, which would be directional. *Boundary:* Descriptive requires an explicit relational claim; a bare co-occurrence in the same sentence is only mentioned.
- **mentioned** Both domains appear near one another but the relationship is not characterized at all. *Choose when:* Two domains named in the same passage with no relational verb. *Not this if:* Any associational, directional, or mechanistic claim. *Boundary:* Prefer mentioned over none only when both domains are present in a shared context rather than in separate isolated sections.
- **none** No relationship between the two domains is articulated anywhere in the review. *Choose when:* Domains handled in separate silos, or one domain absent. *Not this if:* Any co-located relational statement. *Boundary:* If either domain is absent, the pairwise code defaults to none.

integration_triadic

Three-domain integration.

- **mechanistic** A genuinely three-domain mechanism is proposed in which biological, psychological, and social factors act on one another as a system. *Choose when:* A cross-domain loop or cascade involving all three domains; All three domains appear as active parts of one explanation. *Not this if:* Two domains interact while the third is only listed. *Boundary:* Reserve for reasoning where removing any one domain breaks the proposed explanation.

- **descriptive** All three domains are related to the outcome in an integrated narrative, but as a joint description rather than a specified mechanism. *Choose when:* A coherent three-domain account without a stated pathway. *Not this if:* A concrete cross-domain pathway, which would be mechanistic. *Boundary:* Descriptive triadic still requires all three domains to be in genuine relation, not merely co-present.
- **partial** Two domains are integrated with each other while the third is present but only loosely attached. *Choose when:* A strong bio-psych link with social mentioned but not integrated. *Not this if:* All three integrated (descriptive or mechanistic); Only one domain present. *Boundary:* Partial marks a dyadic integration inside a nominally triadic frame.
- **none** No three-domain integration is present; domains stand in parallel or one or more is absent. *Choose when:* Serial mention of separate domains; Single-domain reasoning. *Not this if:* Any genuine multi-domain relational claim. *Boundary:* Serial listing of B, P, and S in turn is none, not descriptive.

integration_claims

One item for every passage in which the paper relates two or three domains to each other. A pairwise field graded above mentioned should have at least one item behind it, and the pipeline checks exactly that. (*extraction list; max 12 items*) *Fields inside each item:*

- **domains_linked** Which domains the passage relates. *Values:* bio_psych, psych_social, bio_social, triadic
- **integration_level** The rung this passage supports. *Values:* mechanistic, directional, descriptive, mentioned, none
- **source_factor_label** The factor doing the influencing, in the paper’s own wording.
- **target_factor_label** The factor being influenced.
- **direction** Whether the influence runs one way or both. *Values:* unidirectional, bidirectional or reciprocal, unspecified
- **mediator_or_moderator** The named intermediate, when the paper gives one.
- **claim_verbatim** The passage, at most 60 words.
- **mechanism_note** The pathway in the coder’s words, empty when none is given.
- **section_located** Where the passage appears. *Values:* abstract, introduction, methods, results, discussion, conclusion, table or figure, other, unclear
- **evidence_basis** What the claim rests on. *Values:* asserted, theorized, empirically_supported, empirically_contested, cited_from_other_work, clinical_observation, other, unclear

Naming both ends of the link is what makes these items usable as an ontology. It lets the synthesis ask which specific factors this literature connects, rather than only how often it connects domains, and it makes two coders’ integration claims comparable as edges rather than as counts.

integration_mechanism_summary

At most 90 words, in the coder’s words: the cross-domain pathways this paper actually proposes. Written as none proposed when the paper proposes none. (*free text*)

F. Typology and Balance

The summary judgments that answer RQ1 at full-text depth. Both are also derived independently by rule, and the two readings are compared (see section J).

overall_balance

Relative emphasis across the three core domains.

- **balanced** No domain dominates; the three are weighted comparably.
- **psych-dominant** Psychological content dominates.
- **bio-dominant** Biological content dominates.
- **social-dominant** Social content dominates.
- **dyadic** Two domains dominate and one is marginal.
- **unclear** Balance cannot be determined.

bps_typology

Full-text BPS operationalization type.

- **true_integrative** Explicit cross-domain causal or mechanistic interaction is central to the review's logic. *Boundary:* Requires at least descriptive triadic integration plus two elaborated domains.
- **multifactorial** Multiple domains covered meaningfully but mostly in parallel. *Boundary:* Distinguished from true_integrative by the absence of a genuine cross-domain link.
- **pseudo_bps** BPS language used but one or more core domains are thin, absent, or tokenistic.
- **rheterical_bps** BPS used mainly as framing or justification without analytic substance.
- **narrow_despite_label** BPS claimed but substantive scope is essentially single-domain.
- **unclear** Type cannot be determined.

The cross-provider test run showed this to be the loosest field in the scheme, both between coders and against the rule-derived typology. It is the field most in need of expert attention.

G. Psychological Concepts, Their Relations, Frameworks, and Measures (RQ3)

The registration asks for concept names, whether definitions are provided and their text, the theoretical frameworks invoked, and the hierarchical and semantic relationships between concepts. All four are coded here as named items rather than as delimited strings, so a concept, its definition, and its relations to other concepts stay attached to one another.

concept_definitions_present

Whether the review defines the psychological constructs it uses.

- **yes** The main constructs are explicitly defined or clearly operationalized.
- **partial** Some constructs defined, others named only.
- **no** Constructs used without any meaning given.

psychological_concepts

Every psychological construct the paper uses, one item each, at the resolution the paper uses it. (*extraction list; max 16 items*) *Fields inside each item:*

- **concept_label** The paper's own term, always.
- **concept_family** The Scheme 5 family it belongs to, when one fits. Empty when none does.
- **definitional_status** What kind of meaning the paper gives it. *Values:* formally_defined, operationalized_only, described_informally, named_only, unclear
- **definition_verbatim** The passage that defines it, at most 60 words, empty when there is none.
- **definition_source** Whose definition it is. *Values:* own definition, cited from other work, taken from an instrument, unattributed, unclear

- **measure_named** The instrument the paper operationalizes it with, when any.
- **factor_role** What the construct does in this paper. *Values:* determinant or risk factor, protective factor, mediator, moderator, outcome, correlate, treatment target, intervention component, contextual condition, descriptive theme, other, unclear
- **section_located** Where the passage appears. *Values:* abstract, introduction, methods, results, discussion, conclusion, table or figure, other, unclear

concept_label and concept_family are both filled and neither replaces the other: the family makes concepts comparable across reviews, the label keeps the distinction between fear of movement during lifting and kinesiophobia.

concept_relations

Every relation the paper draws between two constructs. These are the edges of the concept map the registration asks for. (*extraction list; max 12 items*) *Fields inside each item:*

- **source_concept** The construct the relation starts from.
- **target_concept** The construct it relates to.
- **relation_type** What kind of relation it is. *Values:* is_a_subtype_of, part_of_or_component_of, synonym_or_used_interchangeably, overlapping_or_related, antecedent_or_cause_of, consequence_or_outcome_of, mediates, moderates, measured_by, contrasted_as_distinct_from, conflated_without_comment, other, unclear
- **explicitly_stated** Whether the paper states the relation or merely behaves as though it holds. *Values:* yes, no
- **relation_verbatim** The passage, at most 60 words.
- **section_located** Where the passage appears. *Values:* abstract, introduction, methods, results, discussion, conclusion, table or figure, other, unclear

conflated_without_comment is deliberately available. Silent conflation of two constructs is one of the main findings SQ1 can produce, and it is invisible unless a coder is allowed to record it.

theoretical_frameworks

Every theoretical model or framework the paper invokes. (*extraction list; max 8 items*) *Fields inside each item:*

- **framework_label** The model, in the paper's own wording, with preferred labels available.
- **role** What the paper does with the model. *Values:* organizing framework, tested or modelled, extended or revised, critiqued or rejected, compared with another model, mentioned in passing, other, unclear
- **domains_covered** Which of biological, psychological, and social the model actually spans.
- **attributed_source** The citation the paper credits.
- **framework_verbatim** The passage, at most 60 words.
- **section_located** Where the passage appears. *Values:* abstract, introduction, methods, results, discussion, conclusion, table or figure, other, unclear

instruments

Every measurement or appraisal instrument named. (*extraction list; max 8 items*) *Fields inside each item:*

- **instrument_label** The instrument, with preferred labels available.

- **abbreviation** The abbreviation the paper uses.
- **domain_measured** Which domain the instrument measures in. *Values:* biological, psychological, social, pain or symptom, function or disability, quality of life, multiple domains, methodological quality, other, unclear
- **construct_measured_as_stated** What the paper says the instrument captures, in its own wording.
- **role** What the instrument does in this paper. *Values:* primary outcome, secondary outcome, predictor or covariate, mediator or moderator, screening or classification, developed or validated here, discussed conceptually, critiqued, referenced only, other, unclear
- **instrument_verbatim** The passage, at most 60 words.

What a review measures is the most concrete form its operationalization of the model takes. A paper that claims a biopsychosocial frame and measures only psychological questionnaires has told you something its prose did not.

H. Conceptual Problems (SQ1)

The secondary question of the review. Coded as items rather than as flags, so a problem carries what it is about, which constructs it concerns, and whether the paper noticed it.

conceptual_problems

One item per problem the paper names or displays. An empty list is a legitimate coding. (*extraction list; max 8 items*) *Fields inside each item:*

- **problem_type** The kind of problem. *Values:* vague_definition, tokenistic_bps, missing_social, missing_biology, missing_psychology, mechanistic_absence, construct_overlap, parallel_listing_without_integration, measurement_mismatch, definitional_drift, domain_reductionism, unfalsifiable_or_untestable, other
- **problem_scope** What the problem is about. *Values:* the biopsychosocial model itself, a psychological construct, a biological construct, a social construct, integration between domains, measurement, terminology, scope or coverage, other
- **affected_labels** The constructs or terms the problem concerns.
- **named_by_authors** Whether the paper points the problem out itself, or merely displays it. *Values:* yes, no
- **problem_verbatim** The passage that shows it, which for a displayed problem may be the passage where the gap is visible.
- **note** Anything the fields above cannot hold.

The distinction between a problem the authors name and one they display is the difference between a literature that knows its difficulties and one that does not. Both are coded; only the second is a finding the reviews themselves cannot report.

I. Synthesis Hooks (Free Text)

Deliberately unconstrained fields. They exist so that nuance no controlled vocabulary can hold still reaches the synthesis, and so that the coder never has to force an observation into a field where it does not belong.

key_quotes

The most conceptually load-bearing passages: the ones a reviewer would read first when writing the synthesis. (*extraction list; max 6 items*) *Fields inside each item:*

- **claim_verbatim** The passage, at most 60 words, quotable on its own.
- **claim_type** What kind of claim it is. *Values:* definitional, integrative, operationalizing, critical or problematizing, measurement, theoretical, clinical or applied, other
- **section_located** Where it appears. *Values:* abstract, introduction, methods, results, discussion, conclusion, table or figure, other, unclear
- **why_it_matters** One short sentence on why this passage was selected.

emergent_labels

Every conceptually important term the paper uses that the project vocabularies do not contain: a factor, construct, mechanism, framework, instrument, or population label with no good home on the spine, written exactly as the paper writes it. (*open list; max 12 items; free text*) The review's own error signal. It is how the project ontology learns what it is missing, and the coder is explicitly told to use it generously whenever a spine field had to be left empty.

conceptual_tensions

Contradictions, ambiguities, unresolved debates, and gaps the paper names or displays, including tensions visible inside the paper itself. (*open list; max 5 items; free text*)

additional_observations

Anything else conceptually relevant that no other field captures. One observation per item, as long as it needs to be. (*open list; max 6 items; free text*)

synthesis_note

At most 90 words on what this paper contributes to the question of how the biopsychosocial model is operationalized, and what it does not, written for a reviewer who has not read it. (*free text*)

coding_rationale

At most 40 words justifying the main judgements: the typology, the triadic integration level, and any close call. (*free text*)

J. Derived Fields

Never asked of the coder. Computed from the coded content by fixed rules, so that the same coding always produces the same verdict, in every model and in every run, and so that a change to a rule takes effect without re-coding anything.

presence flags

One yes or no per conceptual element, read off the coded content rather than asked of the coder: BPS usage evidence, a BPS definition, integration evidence, a triadic claim, a named integration edge, biological factors, social factors, other-domain factors, psychological concepts, defined concepts, concept relations, a hierarchical relation, frameworks, instruments, conceptual problems, and domain evidence per domain. (*derived*)

- **yes** The coder returned at least one item of this kind for this paper.
- **no** Nothing of this kind was returned. *Boundary:* An observable fact about the coding, not a judgement: it says this coder recorded no material of this kind in this paper.

These are the variables part of the cross-provider agreement is computed on. Whether two coders both found a framework in a paper has one answer; which label each wrote for it is a different question, answered by the set-overlap metrics on the extraction lists.

coverage and integration depth

coverage_depth per domain, coverage_total, domains_present, pairwise_depth_total, pairwise_depth_max, triadic_depth, and an integration_index between 0 and 1 that averages the normalized pairwise mean with the normalized triadic rung. *(derived)*

ontology breadth

n_subdomains_bio, n_subdomains_psych, n_subdomains_social, n_subdomains_named, n_named_integration, n_emergent_labels, and controlled_label_share. *(derived)* controlled_label_share measures the ontology against the literature, not the coder against the ontology. A low share says this literature is naming things the project vocabularies do not yet carry, and the off-spine labels are the candidate list for extending them.

BPS usage profile

bps_function_set, n_bps_functions, bps_has_substantive_function, and the sections the label appears in, all read off the usage items. *(derived)*

item counts

One count per extraction list, plus n_triadic_claims, n_defined_concepts, n_hierarchical_relations, n_evidence_quotes, n_extracted_items, and n_open_list_entries. *(derived)*

derived_typology

The typology recomputed from coverage and integration by a fixed rule, alongside typology_matches_derived. *(derived)* The typology is the one judgement that is both coded and derived. Comparing the two is the sharpest available test of whether the typology is defined tightly enough to be applied the same way twice. Where they diverge, the codebook is under-specified, not the coder.

conceptual_yield

How much conceptual material the paper actually yielded. A measure of harvest, not of promise. *(derived)*

- **high** Three domains present with at least descriptive triadic integration and three or more integration claims; or four or more claims with two defined concepts or two frameworks; or eight or more distinct subdomains with two or more claims.
- **moderate** Two or more integration claims, or three or more concept relations, or three concepts across two substantive domains.
- **low** Some conceptual material, but thin.
- **minimal** Nothing extracted.

fulltext_eligibility

The post-retrieval verdict. A recommendation for a human adjudicator, not a final decision, and deliberately recall-protecting. *(derived)*

- **include** Eligible and conceptually usable.
- **uncertain** Eligible on the formal criteria but doubtful in substance. *Boundary:* Everything doubtful lands here rather than in exclude. Uncertain is a request for human adjudication.
- **exclude** No biopsychosocial domain content at all, or a single-domain review with no cross-domain claim.

synthesis_priority

Reading order for the later synthesis: core, supporting, background, or not_relevant. *(derived)*

How the Preferred Labels and the Free Text Work Together

Two demands pull against each other in an extraction scheme this large. Comparability wants one label per thing, so that two reviews discussing central sensitization land on the same node whatever they call it. Resolution wants the review’s own words, because a term the ontology cannot hold is a finding about the ontology.

The scheme holds both, by never making them compete for the same field. Each item carries the paper’s own label in one field and, where the project vocabularies apply, an anchor onto the ontology in another: `subdomain_label` sits beside `factor_label`, `concept_family` sits beside `concept_label`. Both are filled. A mapped label never replaces the specific term the paper used.

The mapping itself is conservative. A label is snapped to a canonical one only on an exact match or a whole-token match on one of its lexical variants; anything else is kept as written. The vocabularies are the project’s own: the Scheme 6 subdomain ontology for biological and social factors, the Scheme 5 concept taxonomy for psychological constructs, plus framework, instrument, pain-condition, and attributed-source lists.

Whatever falls outside is wanted rather than discarded. The coder is told explicitly to record the item anyway, leave the anchor empty, and repeat the term in `emergent_labels`. After a run the pipeline reports, per extraction list, how much of what was extracted anchored to the spine and which off-spine labels recur, which is the working list for extending the vocabularies once the expert evaluation is in.

How the Eligibility Verdict Is Derived

The verdict is not asked of the coder. The coder codes what the article contains, and the pipeline then computes eligibility, yield, and priority by fixed rules. Given the same coded content the verdict is fixed, so two coders who read a paper the same way cannot disagree about the verdict, and a disagreement in the output always points back to a disagreement about the paper.

The rule is recall-protecting by design. Only two conditions exclude: no biopsychosocial domain content anywhere in the full text, and a single-domain review with no cross-domain claim. Everything else that looks doubtful becomes uncertain, which is a request for human adjudication rather than a rejection: a source that reads as a primary study, fewer than two domains substantively covered, no biopsychosocial vocabulary with no readable typology, or a typology that cannot be read together with no triadic integration.

This matters because the registration reserves eligibility decisions for human screeners and allows generative AI only as an assistant under human review. The derived verdict is therefore a triage recommendation that tells a reviewer where to look, and the `adjudication_status` field on the coding template is where the human decision is recorded.

Primary Outputs Using This Scheme

- `src/05_test_runs/tests/02_pilot_fulltext/02_model_codings/01_codings/01_one_row_per_paper_and_provider.csv`
- `src/05_test_runs/tests/02_pilot_fulltext/02_model_codings/02_extracted_items/00_all_categories/01_one_row_per_item.csv`
- `src/05_test_runs/tests/02_pilot_fulltext/03_reliability/`
- `src/09_review_stages/04_extraction/forms/stage3_fulltext_coding_template.csv`
- `src/09_review_stages/04_extraction/outputs/stage3_candidate_manifest.csv`